

Unit 9

Computer



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Looking Back

How Alan Turing and His Test Became AI Legend¹

By Dave Roos



(A slate sculpture of Alan Turing by Stephen Kettle at Bletchley Park, near Milton Keynes in Buckinghamshire, England. Turing's portrait hangs on the wall behind.)

1. The Turing Test is legendary in the field of artificial intelligence. First proposed by the visionary British mathematician Alan Turing in a landmark 1950 paper, the test provides a practical (and pretty fun) way to determine if a computer has achieved human levels of intelligence. Turing called it "the imitation game." If a computer—through a text-only chat—can convince a human that it's a real person, then it passes the test. Simple in theory, but nearly impossible in practice.
2. Turing's definition of "computability"—something that a computer can do—is what's known today as an algorithm. Turing was the first to lay out the design framework of a machine that could be programmed to run a series of discrete algorithms in order to achieve a desired task. Other mathematicians and engineers had toyed with calculating machines—most famously Charles Babbage's

¹ The original article is available at [How Alan Turing and His Test Became AI Legend](#) | HowStuffWorks

19th-century analytical engine—but Turing envisioned a device that wasn't limited to solving one kind of problem.

3. "Anything you can describe as an algorithm can be done by one machine," says Andrew Hodges, a mathematics professor at Oxford University and author of "Alan Turing: The Enigma," the inspiration for the Oscar-winning 2014 film "The Imitation Game."

4. "The universal machine is essentially what we mean by a computer now, something on which you can store the instructions and it carries them out," says Hodges. "No one else had formalized that idea."

A Machine With 'States of Mind'

5. From the start, Turing's universal machine was conceived as a very simplified form of artificial intelligence, even though that term wouldn't be coined until 1956. Hodges says that the design of the universal machine was meant to imitate the inner workings of the human mind, a subject that fascinated Turing almost as much as mathematics.

6. In fact, when describing how his universal machine would work, Turing used the term "state of mind" to label the different "read" and "write" functions of the machine. In Turing's conceptual machine, a length of tape is run through a read/write scanner. The tape is inscribed with bits of information represented by symbols. The scanner head can either read the symbols or write new ones according to its "state of mind."

7. "The operation actually performed is determined... by the state of mind of the computer and the observed symbols," wrote Turing in his 1936 paper. "In particular, they determine the state of mind of the computer after the operation is carried out."

8. A decade later, when Turing was leading the stalled British effort to build one

of the first electronic computers in 1946, he also studied neurology and human physiology on the side. The result was an internal paper published for the National Physical Laboratory that modeled how a computer could be programmed to "learn" on its own. Hodges sees it as one of the earliest proposals of what are now called "neural networks," a type of deep machine learning that's at the bleeding edge of artificial intelligence.

The Imitation Game

9. Turing wasn't the only person intrigued by the similarities between human and machine intelligence. A surge of new technologies developed during World War II, including early computers, space satellites and nuclear power, had captured the intellectual and public imagination. "As soon as computers are mentioned at all, people are talking about electronic brains and the possibility of the computer rivaling the brain," says Hodges.

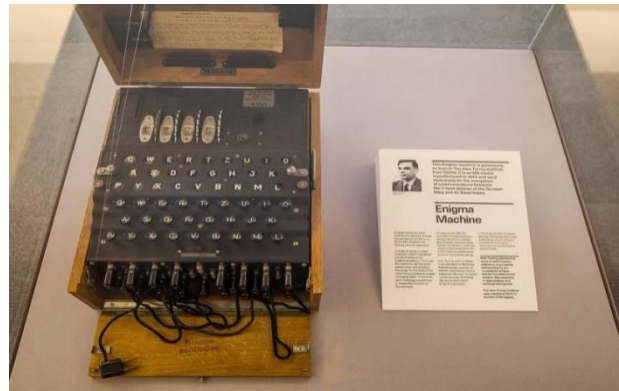
10. The 1948 book "*Cybernetics*" coined the prefix "cyber" and wondered whether it would be possible to "construct a chess-playing machine, and whether this sort of ability represented an essential difference between the potentialities of the machine and the mind." The author, Norbert Wiener, concluded that such a machine "might very well be as good a player as the vast majority of the human race." It was during this era of excitement and nervous speculation about super-intelligent machines that Turing wrote "*Computing Machinery and Intelligence*," what Hodges calls one of the most cited papers in philosophical literature.

11. "I propose to consider the question, 'Can machines think?'" begins Turing. Since the definitions of "machine" and "think" are ambiguous, Turing narrows the scope of the question. For his purposes, the machine must be a "digital computer" and the test of whether or not it can "think" would be answered by the imitation game. It went something like this: There would be three terminals physically separated. Two would have humans as operators (one to be a questioner); the other would have a computer. One human would ask questions to the other human and to the computer via text and, from the answers received from both, determine who was

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"real" and who was computer. A computer was determined to have artificial intelligence (i.e. passed the test) if the questioner could not tell the difference between "man" and "machine."

12. The game, now known as the Turing Test, is only mentioned briefly in the paper and Hodges says that Turing didn't take the details of the test too seriously, publishing different versions in other papers. But Turing did like the playful simplicity of it. "In a way, he was making a drama out of it," says Hodges. It presented this idea [of the possibility of advanced artificial intelligence] in a way that engages people and that ordinary people would make the decision, like a jury in a trial."



(An enigma machine is displayed outside the Alan Turing Institute entrance inside the British Library,

Will a Computer Ever Pass the Turing Test?

13. When the Turing Test was first published in 1950, Turing himself was confident that "intelligent machinery" (as he called it) would be able to win the imitation game within 50 to 100 years. Will his predictions come true?

14. We already have superintelligent computers capable of outwitting the smartest players in other types of games. In 1997, IBM's Deep Blue defeated the reigning chess champion Garry Kasparov, and Watson beat the record-breaking "Jeopardy!" champion Ken Jennings in 2011. But the imitation game raises the bar high on artificial intelligence and no computer has come close to convincing ordinary humans that it's one of them. At least not yet. An annual contest called the Loebner Prize conducts its own Turing Tests on the top chatbots to see if the latest AI software can convince a panel of judges that it's more human than its human

competitors.

15. None of the chatbots have succeeded. The best performer, a conversational chatbot called Mistuku, has only achieved a rating of "33-percent human."

Reading Comprehension

Part A: Inferring and Reasoning

Choose the best inferred statement to each question.

1. Why does the passage mention Charles Babbage's 19th-century analytical engine?

- A. To show that Turing's work was not limited to calculating.
- B. To highlight another historical figure in computing.
- C. To suggest that Turing's ideas were not original.
- D. To emphasize that Babbage's machine was more advanced than Turing's concepts.

2. Why did Hodges think that Turing's work greatly impacted on the modern understanding of computers and algorithms? _____

- A. Turing's work was theoretical and impacted on practical computing.
- B. Turing's algorithms were applicable to mathematical computations.
- C. Turing's work was proven by later advancements in computer science.
- D. Turing formalized the idea of a universal machine, which could perform any algorithm.

3. Why did Turing use the term "state of mind" when describing the functions of his universal machine? _____

- A. He wanted to emphasize the machine's emotional capabilities.
- B. He aimed to draw a parallel between the machine's operations and human cognitive processes.
- C. He was influenced by contemporary psychological theories.
- D. He wanted to confuse his readers with complex terminology.

4. What can be inferred about the significance of Turing's 1936 paper in the field of artificial intelligence? _____

- A. It introduced the concept of neural networks.
- B. It refuted the possibility of machines ever achieving human intelligence.
- C. It laid the groundwork for the design and function of modern computers.
- D. It was primarily a theoretical work with little practical application.

Part B: Generalizing and Summarizing

Fill the blanks with **NO MORE THAN THREE WORDS** in the following summarization of paragraphs 9-15.

Pioneers like Turing and the author of *Cybernetics*, Norbert Wiener, speculated on whether machines could mimic human cognitive abilities and whether there is a crucial distinction between the 5. _____ of them. Turing's seminal paper "Computing Machinery and Intelligence" asked, "Can machines think?" and introduced the imitation game, later known as the Turing Test. This test involved a human interrogator distinguishing between a human and a computer based on their responses, with the computer achieving the status of 6. _____ if the interrogator couldn't tell them apart.

Though Turing predicted that intelligent machinery would pass this test within 7. _____ years, no computer has yet succeeded. While computers like IBM's Deep Blue and Watson have 8. _____ human champions in games like chess and "Jeopardy!," the Turing Test remains an elusive benchmark. The annual Loebner Prize evaluates top chatbots to see if they can pass as human, but the best performer, Mistuku, has only reached a "33-percent human" rating.

Skills

How to compare numbers

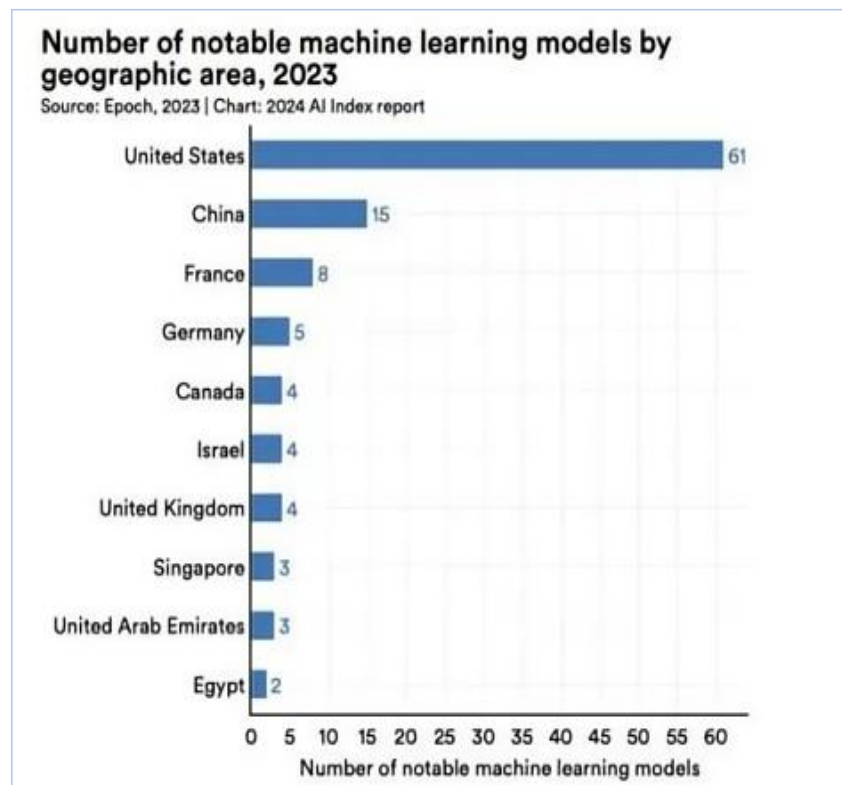
1. Tips of comparing numbers

- ✧ Find the biggest and the smallest number and report .
- ✧ Find the numbers that are about the same and report .
- ✧ You may also do some arithmetic, that is to add some numbers and report the total .
- ✧ You can compare the number you have added with the biggest one or the smallest one .

2. Structure for comparing numbers

- Para. 1 Introduction: paraphrase the prompt or the title of the graph
- Para. 2 An overview: the most striking comparison or trend
- Para. 3 Detail 1: the maximum/the minimum/ the same
- Para. 4 Detail 2 :the maximum/the minimum/ the same

Sample:



Introduction

The bar chart displays the distribution of notable machine learning models in ten countries in 2023.

Overview

There are remarkable differences in the numbers of models developed by these countries and the United States has a huge numerical advantage over other countries in this field.

Detail 1

The U.S. leads significantly with 61 models, far surpassing all other countries. China, though ranking the second place, only has 15 models. Other countries show a noticeable drop in numbers from the top two leaders, ranging from 8 to 3. At the bottom of the list is Egypt, which has just 2 models.

Detail 2

A pronounced disparity exists between the United States and other countries. The United States contributes 56 percent of the total 109 models, while the combined total for remaining countries is 48 models, accounting for only 44 percent.



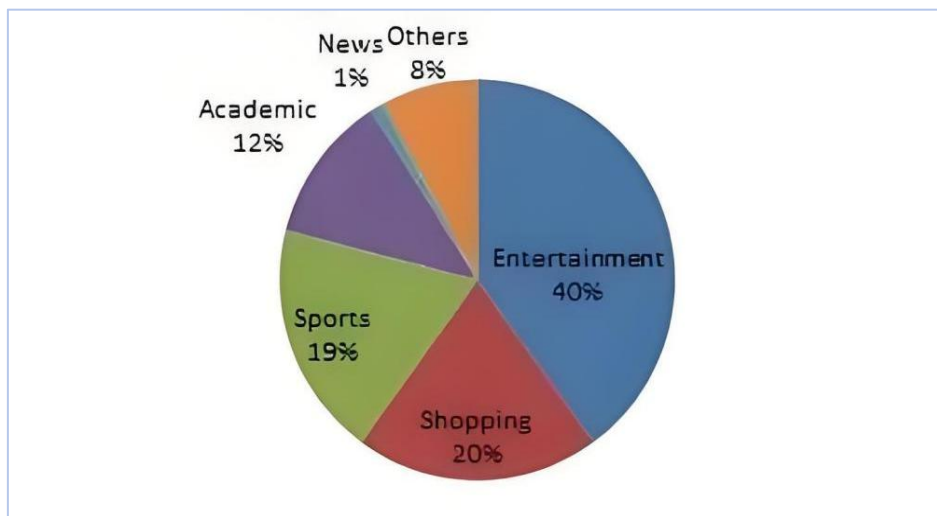
Practice

Please summarize the information and report the main features in the following two graphs.

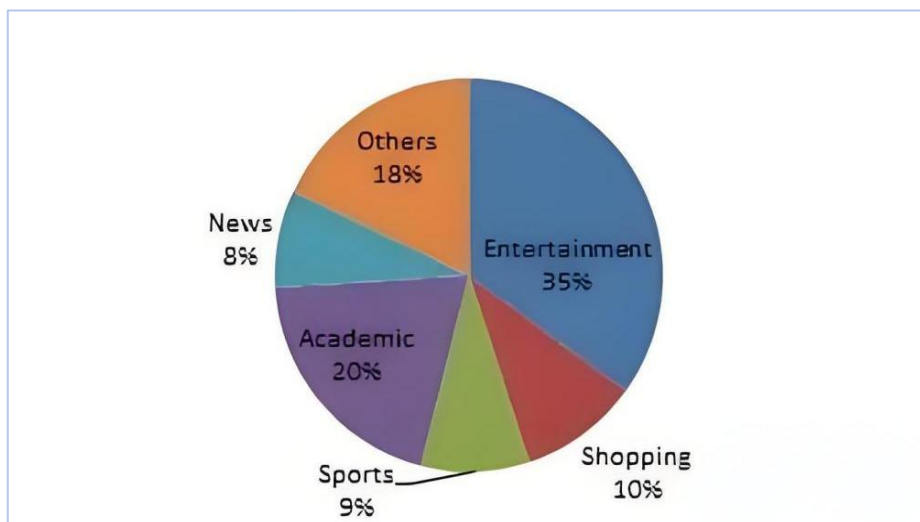
Graph 1

Direction: The pie chart below gives information about the percentage of total time spent on different types of website by both male and female. Summarize the information by selecting and reporting the main features and make comparisons where relevant. Write at least 150 words.

Percentage of total time use on types of website—male



Percentage of total time use on types of website—female



Graph 2

Direction: The table below gives information about percentages of mobile phone owners using various mobile phone features. Summarize the information by selecting and reporting the main features and make comparisons where relevant. Write at least 150 words.

Percentages of mobile phone owners using various mobile phone features

	2006	2008	2010
Make calls	100	100	99
Take photos	66	71	76
Send & receive text messages	73	75	79
Play games	17	42	41
Search the Internet	no data	41	73
Play music	12	18	26
Record video	no data	9	35



Looking Forward

The Future of Computer Science: Four Emerging Technologies and Trends²

1. As a student interested in Computer Science, it is important to stay up to date with the exciting developments and innovations that are shaping the field as it can help you choose a specialization, gain competitive advantage, and adapt to changes. Staying ahead of the curve in the tech world is essential for students and professionals alike as it grants industry-relevant skills and keeps individuals competitive in the field. From well-known and widespread technologies like AI to the rapidly-emerging field of extended reality, students are presented with a diverse array of opportunities to engage and excel in the ever-evolving landscape of computer science.

Artificial Intelligence

2. It is not surprising that Artificial Intelligence (AI) is a leading trend in the computer science field. There are numerous AI applications in computer science growing at a rapid rate, including image recognition, robotics, and even self-modifying code. AI can be a helpful tool in optimizing various processes and tasks across a wide spectrum of industries, such as healthcare, finance, marketing, cybersecurity, and many more. AI-powered systems can analyze vast datasets, identify patterns, and make predictions, enabling businesses to make informed decisions and streamline their workflows.

3. In the engineering field, AI is helpful in developing designs, structures, and

² With technology advancing at an unprecedented rate, the future of computer science holds exciting new developments that are impacting the career field. This blog will explore four top technologies that are trending in the computer science industry.

The original article is available at <https://online.njit.edu/blog-posts/future-computer-science-4-emerging-technologies-and-trends>

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programs as it can create simulation and learn from past data in order to predict outcomes and identify potential problems. Another wonderful example of how AI is taking the future by storm is within healthcare, where AI is making strides in revolutionizing diagnostics through image recognition, aiding in the early detection of diseases.



4. As AI continues to impact various industries, there is an increasing demand for skilled professionals who can design, develop, implement, and maintain AI systems. AI as a required skill for many jobs is expected to increase 65 percent by 2030. Job roles such as Machine Learning Engineer, Data Scientist, and AI Researcher, are growing within different industries.

Cyber security and data privacy

5. With the rapid increase in the use of technology, it is unsurprising that cyber security, the protection of computer systems and networks from malicious attacks, would be on the list of trends in the computer science industry. Computer Engineers, or Information Security Engineers, are tasked with developing and designing secure software and hardware systems. These engineers must identify vulnerabilities in such systems and find and test solutions to ensure its security against cyberthreats. Cybersecurity is a critical component of computer science, especially in this day in age of rapidly advancing technology, as it is essential to safeguarding systems and data from cyberattacks.



6. Information Security occupations are expected to grow 32 percent from 2022 to 2032, with a median wage of about \$110k, making it one of the fastest-growing and

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well-compensated fields in the job market. An educational background in computer science, information systems, and computer engineering can help you break into this lucrative career field.

Extended Reality

7. One of the most exciting fields, growing by a whopping 23.5% in 2023, is Extended Reality, or XR. XR is an umbrella term that includes Virtual Reality (VR), Augmented Reality (AR), and Mixed Reality (MR), each offering a unique blend of digital and physical experiences. Job roles in XR, such as XR Developer, AR/VR Maintenance and Support Technician, and XR Gameplay Engineer, are in high demand as industries increasingly recognize the transformative potential of Extended Reality.

8. Virtual Reality (VR) is a simulated 3D environment that allows users to interact in a completely virtual setting. VR has gained widespread recognition in the entertainment industry, standing out as a pivotal technology in video gaming and beyond. But VR also has useful applications across multiple industries, including healthcare, education, and even marketing.



9. One of technology's most fascinating feats, Augmented Reality (AR), is revolutionizing processes across industries. The AR market is expected to grow nearly 40 percent from 2023 to 2030. AR is a technological innovation that overlays digital information, such as images, videos, and 3D models, onto the real world, creating an immersive experience for users. AR has many useful applications, including in interior design and modeling, where users can visualize the final design, or in medical training where students can learn human anatomy via AR.

headsets.

10. Mixed Reality (MR) combines aspects of both VR and AR, creating an environment where physical and digital objects coexist and interact. This merging opens up new possibilities, particularly in industries like manufacturing and design. For example, engineers can create and test virtual prototypes and train employees in a safe and controlled environment.

Robotic Process Automation

11. The future is here with the transformative power of Robotic Process Automation (RPA). RPA, also known as software robotics, streamlines repetitive business tasks and routines using intelligent automation technologies. RPA involves the use of software robots, bots, to automate repetitive processes, allowing organizations to achieve operational efficiency, reduce costs, and enhance overall productivity. You may be familiar with some of these processes, such as automatic payroll processing or call center automation. But RPA is being increasingly adopted in various businesses and its market is growing exponentially.

12. RPA has many useful applications in businesses, such as data entry, invoice processing, customer service, supply chain management, and financial transactions. As RPA continues to evolve, its integration with other emerging technologies, such as artificial intelligence and machine learning, is expanding its capabilities, making it an exciting field for students and professionals looking to contribute to the ongoing transformation of business processes.

Embracing the future of computer science

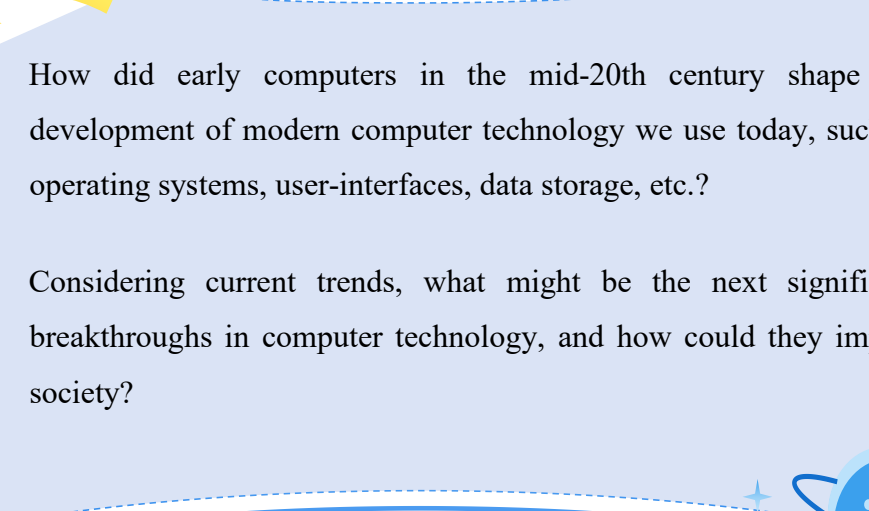
13. The technology field is evolving at a rapid pace that opens a plethora of opportunities, challenges, and careers. Equip you with the agility to adapt to the dynamic changes in the tech landscape. Embrace the future of technology and become a part of innovation.

Translation

当今，人工智能（AI）能优化行业中的流程 and 任务。它可以分析巨大的数据集，识别模式，做出预测，在医疗保健、金融、市场营销、网络安全等领域发挥积极作用。例如，在医疗保健领域，人工智能通过图像识别在诊断病情方面取得进步，有利于人们尽早发现疾病。



Further Discussion

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1. How did early computers in the mid-20th century shape the development of modern computer technology we use today, such as operating systems, user-interfaces, data storage, etc.?
2. Considering current trends, what might be the next significant breakthroughs in computer technology, and how could they impact society?

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